DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-300-05 — Replace VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 (4"-P300-P-1005/B13)

Area	A300 Exchangers and drum	**Task**	VR Valve replacement in kind
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	**E-101** — E-101 shell-and-tube exchanger 36" ID × 192" — E 111'-6" N -25'-0" EL 106'-6"		
Equipment	E-101	**Lines**	4"-P300-P-1005
Isometrics	ISO-4"-P300-P-1005 (generated); owner's drawings ISO-P-1005		
Permits	General work permit		

Reason

Control valve trim worn; replace the valve in kind.

Scope of work

1. Isolate per the isolation list, drain and vent 4"-P300-P-1005 between the isolations.
2. Unbolt and remove VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 (4"-P300-P-1005/B13) at E 33'-6" N -15'-7-15/16" EL 121'-0".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 of the same size, class and face-to-face (CONTROL4-P300), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 4"-P300-P-1005/B13, 4"-P300-P-1005/B15, 4"-P300-P-1005/B12

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1005	4"-P300-P-1005	4"-P300-P-1005	4	0 error(s), 5 warning(s) — see below

ISO-P-1005 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1005	1	warning	weld 4"-P300-P-1005/W11 (E 109'-11/16" N -4'-0" EL 116'-6", sheet 1) has no W11 label
ISO-P-1005	3	warning	W18 (4"-P300-P-1005/W18, sheet 3) is labelled again on sheet 2; it is 223.3 in from the nearest break
ISO-P-1005	3	warning	weld 4"-P300-P-1005/W19 (E 34'-11/16" N -2'-0" EL 116'-6", sheet 3) has no W19 label
ISO-P-1005	4	warning	weld 4"-P300-P-1005/W21 (E 33'-6" N -2'-0" EL 117'-11 5/16", sheet 4) has no W21 label
ISO-P-1005	2	warning	branch weld 3/4"-P300-I-7005/W1 (E 109'-0" N -2'-0" EL 116'-9 5/16", sheet 2) has no W30 label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
E-101	equipment	E-101 shell-and-tube exchanger 36" ID × 192"		E 111'-6" N -25'-0" EL 106'-6"
4"-P300-P-1005/B13	item	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300	4"-P300-P-1005	E 33'-6" N -15'-7-15/16" EL 121'-0"
4"-P300-P-1005/B15	item	GASKET 4" P300 COMPRESSED SHEET	4"-P300-P-1005	E 33'-6" N -14'-11-11/16" EL 121'-0"
4"-P300-P-1005/B12	item	GASKET 4" P300 COMPRESSED SHEET	4"-P300-P-1005	E 33'-6" N -16'-4-1/8" EL 121'-0"
4"-P300-P-1005/B16	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1005	E 33'-6" N -14'-4" EL 121'-0"
4"-P300-P-1005/B10	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1005	E 33'-6" N -16'-11-13/16" EL 121'-0"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P300-P-1005/B16	4"-P300-P-1005	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1005/B16)	E 33'-6" N -14'-4" EL 121'-0"		

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
2	blind	4"-P300-P-1005/B13	4"-P300-P-1005	Fit a line blind between 4"-P300-P-1005/B16 and VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 (4"-P300-P-1005/B13), work side	E 33'-6" N -16'-4-1/16" EL 121'-0"		
3	valve	4"-P300-P-1005/B10	4"-P300-P-1005	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1005/B10)	E 33'-6" N -16'-11-13/16" EL 121'-0"		
4	blind	4"-P300-P-1005/B13	4"-P300-P-1005	Fit a line blind between 4"-P300-P-1005/B10 and VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 (4"-P300-P-1005/B13), work side	E 33'-6" N -16'-4-1/16" EL 121'-0"		

Parts

Description	Qty	Replaces
VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300	1	4"-P300-P-1005/B13
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1005/B15
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1005/B17
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1005/B12
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1005/B14

Inspections

Id	Inspection	Criteria	Basis
WO-300-05-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
4"-P300-P-1005/B15	4 × 1" studs (4"-P300-P-1005/B17)	35,343	3	
4"-P300-P-1005/B12	4 × 1" studs (4"-P300-P-1005/B14)	35,343	3	

Test — Service leak test

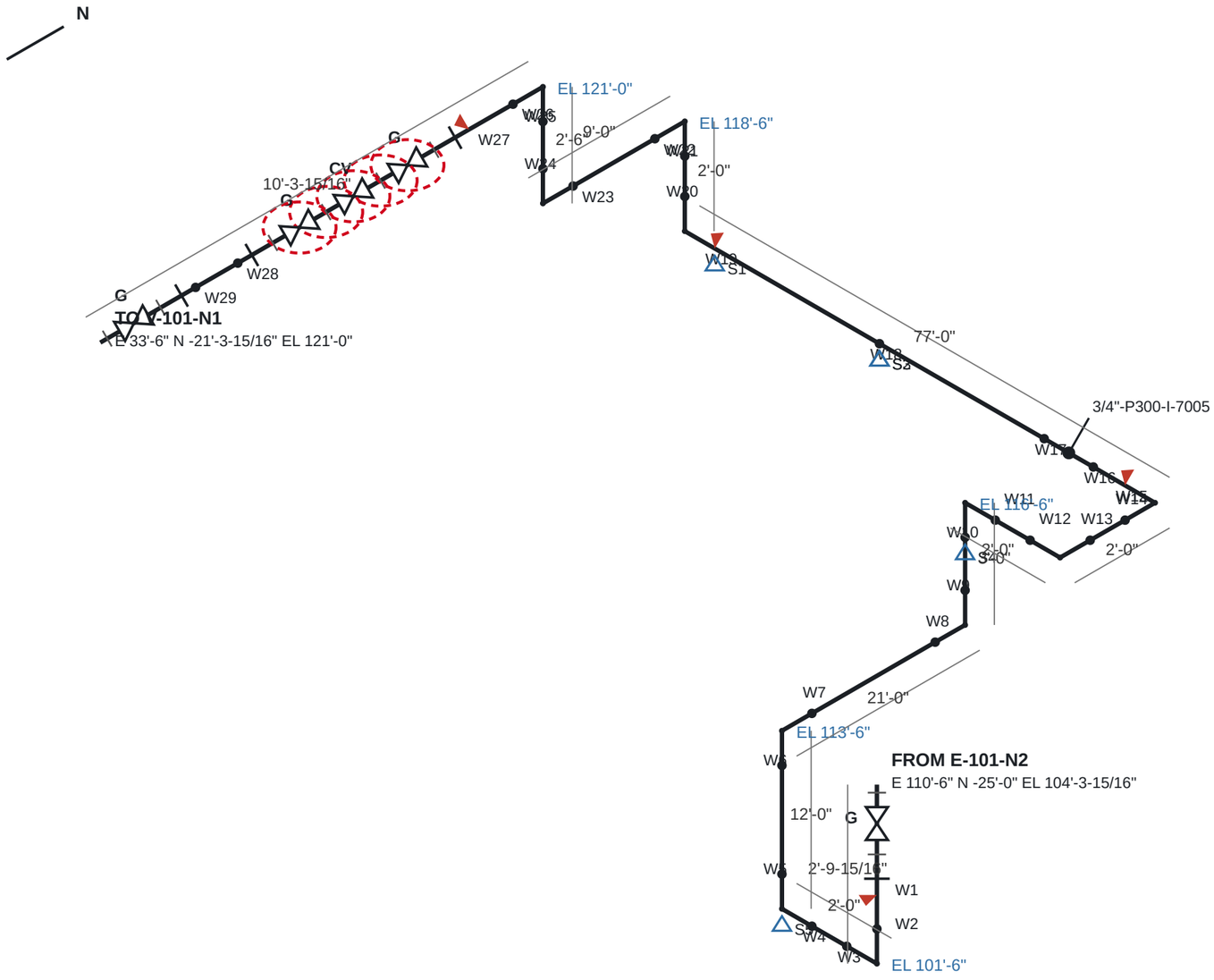
Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 4"-P300-P-1005/B13 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	CONTROL4-P300	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P300-P-1005 — Replace VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 (4"-P300-P-1005/B13)



BILL OF MATERIALS

1	GASKET 4" P300 COMPRESSED SHEET	8
2	VALVE, GATE, WEDGE, OS&Y 4" P300	4
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	8
4	FLANGE WELD-NECK 4" P300 RAISED FACE	4
5	PIPE 4" P300 BEVELLED ENDS	124'-6-1/2"
6	ELBOW 90° R = 5/3 OD 4" P300	11
7	TEE REDUCING 4" × 3/4" P300	1
8	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300	1

WELDS 29 (4 FIELD) · SUPPORTS 5 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15F W16 W17 W18 W19F W20
W21 W22 W23 W24 W25 W26 W27F W28 W29

4"-P300-P-1005

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
E-101-N2 → V-101-N1
PIPE 124'-6-1/2" · WEIGHT 1974 lb

DWG ISO-4"-P300-P-1005 REV 0 WO-300-05

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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